

A3: LLM Language Application & Evaluation

Overview & Goal

Option 1. LLM application. In this assignment, you'll have the opportunity to design a language-based application or tackle an existing task in NLP using a Large Language Model (LLM). As this is quite open ended, creativity is encouraged. It may be helpful to take inspiration from recent research or shared tasks in NLP (see *Resources* below). Alternatively, you may be inspired to respond to a need in your university or community, or simply explore an idea or method you've always wanted to learn more about.

As you develop your project, be sure to incorporate methods and ideas from our course. Your methodology should reflect a thoughtful engagement with how LLMs work, their strengths, and their limitations. The choices you make—about the prompting and language used, model chosen, data, and human-LLM interaction—should be informed by theory and existing practice and will be assessed in accord.

Essential to a successful project is how you evaluate the use of an LLM. As we have seen, while LLMs can be impressive in their linguistic performance, they are not always consistent or reliable. Part of your task is to design a way to assess your system's reliability and reproducibility, and to reflect critically on where it succeeds and where it might fall short.

Ideas from previous submissions

- ★ Basic translations into a dialect or low-resource language, or help for language learning with this language or dialect.
- ★ Recipe app to walk users through cooking steps (can be done for any instructional task), or e.g. adapt a recipe to certain dietary restrictions (e.g. vegetarian).
- ★ Quiz app to review course materials in interactive ways.
- ★ Naming newly discovered plants or animals, or create new comic characters, based on certain visual characteristics described through language.
- ★ Lyric generator based on musical genre, artist, language, topic, etc.
- ★ Word similarity game based on word vectors. Play here: <https://word-void.onrender.com/>

(If you choose to work on Option 2 or 3, please discuss with Lucia)

Option 2. Build a small language model. “Language models (LMs) are powerful tools for natural language processing, but they often struggle to produce coherent and fluent text when they are small.” To tackle this challenge, read the paper on [TinyStories](#), and build a small language model of your own for a specific task – for example, developing a model for a specific language or genre. There are many tutorials online you can follow, as well as [this notebook](#).

Option 3. Build a cognitively plausible language model. Huge effort has been put into optimizing LM pretraining at massive scales in the last several years. To counter this, many researchers now focus their efforts on optimizing pretraining given data limitations inspired by human development. Here, you will build a cognitively plausible model following the instructions of the [BabyLM Challenge](#). All information is available on the website, and you can submit your results for publication if you choose!

**As usual, if you have questions about the structure of the assignment, where to find additional datasets or resources, or would simply like to talk through ideas please email Lucia or ask your TA.*

Submission Guidelines

For this assignment, you may work individually or in a group (up to three people) of your choice. Each individual should submit their own assignment in a .zip file (NAMES_A3_NLP2026) with:

1. Your code. This can be in .py files, a Jupyter notebook, Colab, or in a GitHub repo. All code should be well-documented and be able to be run by someone else.
2. A PDF write-up following [this template](#) (which you must copy) that describes:
 - a. **Motivation:** What contribution are you making with your project?
 - b. **Linguistic Relevance:** What linguistic challenges are addressed in this project?
 - c. **Methodology:** How do you create your app or solve your task? This should include details on which LLM you choose and why, LLM settings, design decisions, etc. If you are working with data, be sure to explain in detail the dataset you are working with: If it is an existing dataset, what characteristics does it have? If you are generating a dataset with an LLM, how do you do this and verify its reliability.
 - d. **Results/Evaluation:** What does your model produce? How do you test for consistency or fairness? Do you evaluate accuracy, usability, trustworthiness, something else? You might use benchmarks, user studies, or error analysis—whatever best fits your project.
 - e. **Final Thoughts:** What did you learn from this project? How would you improve it in future work?

Deadline Friday, May 8 (by midnight)

Grading This assignment is optional. Students may earn up to 1 additional point on their overall assignment grade. As this is significant, grading will be deliberate and rigorous. Final grades will be based on (i) code correctness and difficulty; (ii) creativity of idea; (ii) your write-up with analysis of why your app is linguistically meaningful, including screenshots if applicable.

Resources

LLMs: Depending on how much compute you have access to, you may need to use a smaller model. [HuggingFace](#) is a great resource for finding models, and [here](#) you can find smaller open-source ones. If you are interested in using a VLM, you are welcome to.

Prompting: [Here](#) is a great overview of LLM settings and methods.

Shared Tasks: As LLMs have become a staple of much recent research, you may choose to look at recent papers, datasets, or shared tasks for inspiration. Searching for “LLM NLP Shared Task” (or the like) should offer many options. The [ACL Anthology](#) is a great resource for additional, peer-reviewed research on LLMs. Here are some curated tasks for you, too:

- ★ Stanford assignment leveraging LLMs to create task-oriented dialogue agents grounded on knowledge corpus: [link](#)
- ★ Shared task on LLM financial abilities: [link](#)
- ★ LLMs as educational tutors: [link](#)
- ★ LMs trained on similar data as human child: [link](#)
- ★ English bias of LLMs in Dutch and French (can extend to other languages): [link](#)
- ★ Propaganda detection in Arabic: [link](#)